

Ubuntu Linux Administration: Networking

Configuring IP Addresses in Ubuntu Linux



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Overview



Ubuntu Networking

Course Overview

- IP configuration
- Hostname resolution
- NTP client
- Routing between networks
- Firewalling with IPtables and UFW

In this Module

- Using the ip command
- Configuring transient IP addresses
- Configuring persistent IP configuration with netplan





Lab Setup

Ubuntu Linux Administration: Essential Commands

Andrew Mallett





206.132.41.231
from 206.132.41.231: 1000
from 206.132.41.231: 1000
from 206.132.41.231: 1000
from 206.132.41.231: 1000
from 206.132.41.231: 1000
from 206.132.41.231: 1000

IP Addresses

Transient

- deprecated: ifconfig
- replaced with: ip



```
$ sudo apt install -y net-tools  
$ ifconfig  
$ ip address show  
$ sudo ip addr add 192.168.1.1/24 dev enp0s8
```

Ip Replaces Ifconfig

Many administrators have become used to the ifconfig command for many years. By default, it is not installed in Ubuntu 20.04 but can be added via the net-tools package. In reality, ifconfig is deprecated in favor of the new command ip. This is included, out-of-the-box in most if not all distributions.

Demo



Working with both Ubuntu1 and Ubuntu2:

- Read IP address configuration with ip and ifconfig
- Add a new address to both systems
- Test connectivity





...41.231
...132.41.231 (206.132.41.231)
from 206.132.41.231: 100.00
from 206.132.41.231: 100.00
from 206.132.41.231: 100.00
from 206.132.41.231: 100.00
from 206.132.41.231: 100.00
from 206.132.41.231: 100.00
from 206.132.41.231: 100.00

IP Addresses

Persistent

- managed using netplan
- /etc/netplan/*.yaml
- /usr/sbin/netplan



```
$ netplan info
$ netplan ip leases enp0s3
$ sudo vim /etc/netplan/50-vagrant.yaml
---
network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s8:
      addresses:
        - 192.168.56.101/24
        - 192.168.2.201/24
$ sudo netplan try
```

Persisting the Configuration

Ubuntu 20.04 uses netplan to manage networking configuration. The files are written in YAML. We can add new IP addresses to netplan to persist the network configuration, even with the ability to have an automated rollback in the event of a misconfiguration

YAML files are whitespace
sensitive indicating data
structures



Demo



Working with both systems we persist IP address configuration:

- netplan and DHCP leases
- netplan and rollback
- netplan apply



Summary



IP Address Configuration in Ubuntu

- ip
- ifconfig
- netplan
- YAML



Configuring Hostname and Hostname Resolution in Ubuntu Linux

